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I am currently conducting my PhD study in the Department of Bioenvironmental Systems Engineering at National Taiwan University, Taiwan. I hold undergraduate and Master's degrees from the Department of Earth Sciences at National Central University (NCU) and the Graduate Institute of Hydrological and Oceanic Sciences, NCU, respectively.

My current research focuses on soil water dynamics at the Darcy and pore scales. I have also participated in research projects associated with irrigation and water resource management under the impacts of climate change. Additionally, as a graduate of the Department of Earth Sciences and the Graduate Institute of Hydrological and Oceanic Sciences, I possess a solid foundation in geophysics and hydrology.

Experience

2018/9-2020/7 Research Assistant, Department of Bioenvironmental Systems Engineering, National Taiwan University, Taipei, Taiwan

Education

2020-present Ph.D. Student, Department of Bioenvironmental Systems Engineering, National Taiwan University, Taipei, Taiwan
2015-2017 M.S., Graduate Institute of Hydrological and Oceanic Sciences, National Central University, Taoyuan, Taiwan
2011-2015 B.S., Department of Earth Science, National Central University, Taoyuan, Taiwan

Peer-reviewed Journal Publications

1. **Huang, Q.-Z.**, Hsu, S.-Y.*; Li M.-H. (2026) "Predicting future exceedance probability of streamflow using the long-term weather outlook and logistic regression: a case study in Northern Taiwan" *Terrestrial, Atmospheric and Oceanic Sciences*, 37(4) <https://doi.org/10.1007/s44195-025-00120-x>
2. **Huang, Q.-Z.**, Hsu, S.-Y.*, Hu, J., & Chang, Y.-C. (2024) "Observations of Electrical Resistivity Tomography and Groundwater Level Under an Irrigated Terraced Paddy Field in Northern Taiwan." *Paddy and Water Environment*. <https://doi.org/10.1007/s10333-024-00979-5>
3. Wang, Y.-L., **Huang, Q.-Z.**, & Hsu, S.-Y.* (2023) "The Displacement of the Resident Wetting Fluid by the Invading Wetting Fluid in Porous Media Using Direct Numerical Simulation." *Water*, 15(14):2636. <https://doi.org/10.3390/w15142636>
4. **Huang, Q.-Z.**, Huang, J.-C., Tsao, C.-W., & Hsu, S.-Y.* (2023). "Pore Doublet Micromodel Experiments of Evaporation Influence on Pre-Event Water Entrapment and Pre-Event and Event Water Interaction." *Advances in Water Resources*, 177, 104464. <https://doi.org/10.1016/j.advwatres.2023.104464>
5. **Huang, Q.-Z.**, Hilpert, M., Tsai, Y.-Z., & Hsu, S.-Y.* (2023) "Investigation and Parameterization of the Ponding Height Effect on Dynamic Suction Head Estimation in the Green-Ampt Model." *Journal of Hydrology*, 2023, 621, 129524. <https://doi.org/10.1016/j.jhydrol.2023.129524>
6. Lo, J.-H., **Huang, Q.-Z.**, Hsu, S.-Y.*, Tsai, Y.-Z., & Lin, H.-Y. (2022). "Evaluating Spatial-Temporal Clogging Evolution in a Meso-Scale Lysimeter." *Land*, 11(9):1518. <https://doi.org/10.3390/land11091518>
7. Shih, Y.-H., Hsu, S.-Y.*, **Huang, Q.-Z.**, Lamorski, K., Hu, M.-C., Tsao, C.-W., Sławiński, C., & Shokri, N. (2021). "Euler Characteristic During Drying of Porous Media." *Drying Technology*, 40(4), 781-795. <https://doi.org/10.1080/07373937.2021.2007946>
8. Tsao, C.-W., **Huang, Q.-Z.**, You, C.-Y., Hilpert, M., Hsu, S.-Y.*, Lamorski, K., Chang, L.-C., & Sławiński, C. (2021). "The Effect of Channel Aspect Ratio on Air Entrapment During Imbibition in Soil-on-a-Chip Micromodels with 2D and 2.5 D Pore Structures." *Lab on a Chip*, 21(2), 385-396. <https://doi.org/10.1039/D0LC01029D>

9. Tsai, Y.-Z., Cheng, M.-L., **Huang, Q.-Z.**, Lo, W.-C., Li, M.-H., & Hsu, S.-Y.* (2020). "Effect of effective saturation and ceramic cup properties on the response time of tensiometers." *Journal of Hydrology*, 582, 124445. <https://doi.org/10.1016/j.jhydrol.2019.124445>

International Conference Publications

1. Hsu, S.-Y., Zhan, K.-X., **Huang, Q.-Z.**, Chang, T.-J., Tsao, C.-W. (2023). "Using Capillary Tube and Pore Doublet Micromodel to Investigate the Effect of Entrapped Gas Phase on the Mass Transport of Carbon Dioxide in Unsaturated Zone, Solute". *AGU Fall Meeting*, 2023.
2. Lee, J.-Y., **Huang, Q.-Z.**, Chen, C.-C., Liu, T.-S., Hsu, S.-Y., Huang J.-C., Peng T.R. (2023). "Dynamics Dependence on Soil Water Ages in a Maize Farm". *AGU Fall Meeting*, 2023
3. **Huang, Q.-Z.**, Hsu, S.-Y., Hu, J. (2023). "Electrical Resistivity Tomography and Groundwater Level below the Irrigated Terraced Paddy Field in Northern Taiwan". *AGU Fall Meeting*, 2023.
4. **Huang, Q.-Z.**, Hsu, S.-Y., Hu J. (2023). "Electrical Resistivity Tomography and Groundwater Level Below the Irrigated Terraced Paddy Field in Northern Taiwan". *25th International Congress on Irrigation and Drainage*, 2023.
5. **Huang, Q.-Z.**, Hsu, S.-Y. (2022). "Exceedance Probability of Streamflow Based on the Long-term Weather Outlook". *Asia Oceania Geosciences Society*, 2022.
6. **Huang, Q.-Z.**, Hsu, S.-Y. (2021). "Visualizing the Subsurface Flow in Terraced Paddy Fields by Using Geoelectric Resistivity Tomography--a Case Study". *INWEPF-PAWEES International Conference*, 2021.
7. **Huang, Q.-Z.**, Huang, J.-C., Hsu, S.-Y., Tsao, C.-W. (2019). "The effect of evaporation on the interactions between the trapped and invading wetting phase fluids in the pore doublet micromodel". *AGU Fall Meeting*, 2019.
8. Lin, H.-H., Hsu, S.-Y., **Huang, Q.-Z.**, Tsao, C.-W., Chang, L.-C., Sławiński, C., Lamorski, K. (2019). "Evaporation effect on pore-scale concentration changes and interplay between the resident and invading liquids in micromodels". *AGU Fall Meeting*, 2019.
9. Hsu, S.-Y., Tsai, Y.-Z., **Huang, Q.-Z.**, Cheng, M.-L., Lo, W.-C., Li, M.-H. (2019). "Effect of the effective saturation and ceramic cup properties on the response time of tensiometer". *AGU Fall Meeting*, 2019.
10. **Huang, Q.-Z.**, Huang J.-C., Hsu S.-Y., Tsao C.-W. (2019). "The Study of the Effect of the Evaporating on New and Old Water Interplay by Using Pore Doublet Micromodel Experiment". *INWEPF-PAWEES International Conference*, 2019.
11. Lin, H.-H., Hsu, S.-Y., Tsao, C.-W., Shih, Y.-H., **Huang, Q.-Z.** (2019). "Investigation of the Effect of Evaporation on the New and Old Water Interplay in Porous Media by Using Micromodel Experiment". *INWEPF-PAWEES International Conference*, 2019.
12. Lin, G.-Z., **Huang, Q.-Z.**, Hsu, S.-Y., Lee, T.-Y., (2019). Evaluate the impact of extreme weather on the water resource and crop growth in Northern Taiwan. *INWEPF-PAWEES International Conference*, 2019.